Climate-Finance Scenario, Portfolio and Validation Methodology

A traceable research architecture for strategic Climate VaR and tactical event risk

Prepared for	Jennison client review
Version	21 July 2026
Model scope	U.S.-listed equity and ETF research prototype
Primary use	Conditional scenario repricing, portfolio risk budgeting, event overlays, and walk-forward validation

Central design rule Climate information is not assumed to be alpha. It enters strategic allocation through conditional cash-flow loss, scenario penalties, explicit physical and transition budgets, and a separately validated tactical event overlay.

Research prototype only. Not investment advice, a regulatory capital model, a probability forecast, or a substitute for issuer due diligence. The supplied Bloomberg methodology was used as a conceptual reference; no Bloomberg proprietary coefficients or datasets are reproduced.

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1 Executive decision summary

The model is designed to answer four distinct questions without collapsing them into one opaque score:

1. **Strategic value:** how would an issuer or portfolio be valued today if markets repriced from a chosen reference climate path to a target path?
2. **Tactical event risk:** which holdings are exposed to today's hazard observations, and what small, funded overlay follows from that exposure?
3. **Portfolio construction:** what allocation maximizes a market-derived expected-return proxy while controlling covariance risk and conditional climate losses?
4. **Evidence:** does adding climate information improve return, drawdown, or tail behavior relative to the same market-only process?

The architecture follows the strongest elements of scenario-analysis practice: a target-versus-reference comparison, annual cash-flow paths, country-to-sector-to-issuer transmission, firm-specific discount rates, and as-of-date repricing. It extends that structure with a separate NWS event clock, transparent model-risk bands, a robust portfolio objective, and a climate-on versus climate-off validation design.

What the client should focus on The most decision-relevant outputs are the scenario-conditioned value impact, the cash-flow attribution by driver, the portfolio's worst conditional scenario loss, the active risk versus the market-only portfolio, and the incremental climate value with uncertainty. A high dashboard score by itself is not an investment conclusion.

1.1 What changed in this model revision

Issue	Previous weakness	Revised treatment
Transition risk	Atmospheric CO_2 could be mistaken for policy pressure	CO_2 is context only; transition risk is decomposed into carbon cost, demand, energy cost, transition CapEx, and pass-through
GDP transmission	One macro score	Country GDP, sector demand, and supply-chain channels are separate and traceable
Physical risk	Alert counts could look like probabilities	Event loss is hazard severity × location exposure × vulnerability ×(1 − resilience)
Time horizon	Recent prices could contaminate long-term climate valuation	Strategic 5–30 year model and tactical 1–20 trading-day overlay are separate clocks
Tail metric	Sparse scenarios could be labeled like Expected Shortfall	Conditional scenario range, worst case, and model-ensemble tail are shown; no regulatory ES claim
Portfolio	Climate score could be interpreted as unvalidated alpha	Climate enters as loss penalties and risk budgets; expected-return input remains market-derived
Backtest	One result could be mistaken for proof	Climate-on versus market-only ablation, costs, bootstrap interval, permutation test, and explicit data limitations

2 System architecture and two clocks

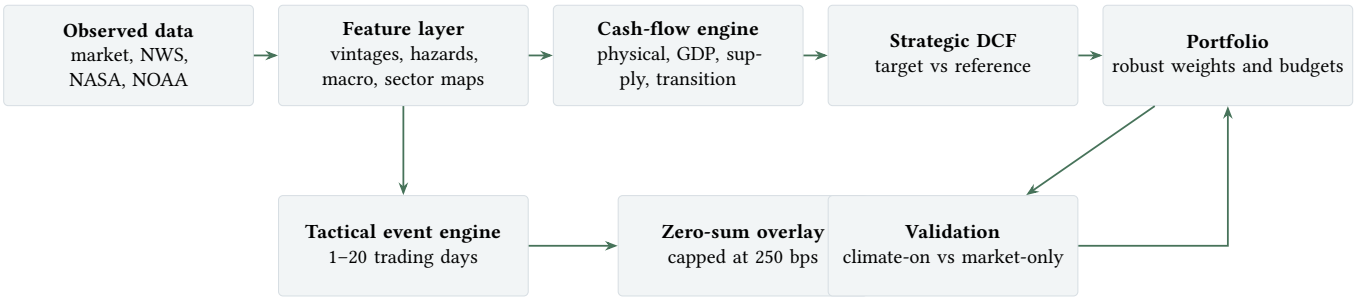


Figure 1: Closed-loop architecture. Strategic valuation and tactical event response share source lineage but do not share a time clock.

The final weight is

$$w_t = w_t^{\text{Strategic}} + \Delta w_t^{\text{Event}}, \qquad \mathbf{1}^\top \Delta w_t^{\text{Event}} = 0, \qquad |\Delta w_{i,t}^{\text{Event}}| \le 250 \text{ bps}. \tag{1}$$

The strategic model uses annual scenario paths and one-year daily market history for covariance and the market return proxy. Current NWS alerts do not alter strategic weights. The event overlay can update daily, but it must fund itself, respect position and sector caps, and remain small relative to the strategic allocation.

3 Data architecture and lineage

3.1 Observed, calibrated, and missing

Source	Model role	Current status	Control and interpretation
NWS active alerts	Tactical acute hazard severity	**Implemented**	Used as a current event nowcast, never as a long-run disaster probability
NASA GISTEMP	Chronic climate state	**Implemented**	Current warming context and chronic-risk calibration
NOAA/NCEI disasters	Historical physical-loss context	**Implemented**	Disaster frequency and loss history; not an issuer damage estimate
NOAA GML CO_2	Climate-state context	**Implemented**	Displayed for context and excluded from transition risk
NGFS scenario paths	GDP, inflation, energy, carbon, physical pathways	**Calibrated proxy**	Transparent scenario calibrations are operational; production-grade annual ingestion and version pinning remain required
BEA GDP by Industry	Sector value added and GDP sensitivity	**Calibrated proxy**	Model contains a BEA-style sector mapping; direct production ingestion remains source-ready
BEA Input-Output	Direct and indirect supply-chain transmission	**Calibrated proxy**	Supply beta is explicit; full requirements-table ingestion remains source-ready
FRED/ALFRED	Point-in-time macro vintages	**Data gap**	Required before macro backtests can be called fully point-in-time
EIA and EPA eGRID	Energy price, generation mix, emissions intensity	**Calibrated proxy**	Sector calibrations are explicit; issuer/facility joins remain source-ready
SEC XBRL	Revenue, cost, CapEx, capital structure	**Calibrated proxy**	WACC uses market beta and sector capital-structure fallback; direct issuer ingestion remains source-ready

Source	Model role	Current status	Control and interpretation
Facility coordinates	Bottom-up hazard exposure	**Data gap**	A documented issuer-facility schema exists; absent records are labeled as sector fallback
Supplier/customer graph	Indirect transition and interruption risk	**Data gap**	Needed for issuer-level second-order exposure

Official sources are chosen because they support traceability. BEA requirements tables describe both direct and indirect inputs across industries; ALFRED exposes historical vintages; NASA NEX-GDDP-CMIP6 provides bias-corrected downscaled daily projections; EPA eGRID describes power-sector emissions and resource mix; EIA provides energy prices and generation data; and SEC APIs provide machine-readable filings. These sources improve auditability but do not remove model risk.

3.2 Point-in-time rule

For a signal formed at date t, every feature must satisfy

$$\text{availability_date}(x) \leq t. \tag{2}$$

Annual climate features use the latest completed calendar-year vintage and expanding-window normalization. Market features use observations strictly prior to the next-day return. Production validation additionally requires historical index membership, delisted securities, filing timestamps, archived NWS/NOAA observations, and ALFRED macro vintages.

4 Strategic model: scenario to firm value

4.1 Target versus reference

Let s denote the target scenario and r the reference. The model evaluates the immediate, as-of-date valuation change implied by replacing the reference path with the target path. It does not assign a probability to either path:

$$\Delta X_{s,t} = X_{s,t} - X_{r,t}. \tag{3}$$

This aligns with the scenario-conditioned repricing principle in the supplied Bloomberg methodology. It is a sensitivity analysis, not a forecast.

4.2 Country GDP, sector demand, and supply chain

The revenue-growth channel is decomposed rather than hidden inside one macro score:

$$\Delta g^{\text{Revenue}}_{i,s,t} = \beta_i^{GDP}\Delta GDP_{c,s,t} + \beta_i^{Demand}\Delta Demand_{k,s,t} + \beta_i^{Supply}\Delta SupplyFactor_{s,t}. \tag{4}$$

The implementation uses a country path, BEA-style sector GDP and input-output betas, and an issuer earnings-sensitivity fallback. Revenue geography and actual supplier/customer weights are explicitly marked missing when they are unavailable. Country, sector, and supply-chain terms remain separate in both the DCF and the Shapley attribution, preventing the same demand effect from being counted twice.

4.3 Transition cash-flow channels

Atmospheric CO_2 concentration is a climate-state observation, not a policy instrument. Transition risk is therefore defined through financially interpretable channels:

$$\begin{aligned} TR_{i,s,t} = {} & CarbonCost_{i,s,t} + DemandShift_{i,s,t} + EnergyCost_{i,s,t} \\ & + TransitionCapex_{i,s,t} - PassThrough_{i,s,t}. \end{aligned} \tag{5}$$

Each term maps to an issuer cash-flow line:

- *CarbonCost*: emissions intensity × scenario carbon-price shock;
- *DemandShift*: product/sector demand path × demand elasticity;
- *EnergyCost*: energy intensity × scenario energy-price shock;
- *TransitionCapex*: required investment intensity × scenario adjustment path;
- *PassThrough*: the share of carbon and energy costs recoverable through prices.

Positive values reduce FCFF; pass-through is recorded as a negative loss. Current sector coefficients are disclosed fallbacks until issuer-level NGFS, EIA, eGRID, and SEC joins are loaded.

4.4 Top-down physical risk

Country physical losses are conserved when downscaled to sectors. For risk type $q \in \{\text{acute}, \text{chronic}\}$,

$$\Delta Y^q_{j,t} = \frac{\alpha^q_j}{\sum_k \omega_k \alpha^q_k} \Delta Y^q_{c,t}, \tag{6}$$

where ω_k is sector k's value-added share. The weighted sector multiplier reconciles to one. Firm-level differentiation is bounded:

$$\Delta Y^q_{i,j,t} = \text{clip}\left(\frac{\alpha^q_{i,j}}{\tilde{\alpha}^q_j}, a_{\min}, a_{\max}\right) \Delta Y^q_{j,t}. \tag{7}$$

Fixed-asset intensity proxies acute vulnerability and labor intensity proxies chronic vulnerability. Missing issuer observations use labeled sector medians.

4.5 Bottom-up event loss

The tactical physical engine uses a different equation:

$$Loss_{i,h,t} = Hazard_{h,l,t} \times Exposure_{i,l} \times Vulnerability_{i,h} \times (1 - Resilience_i). \tag{8}$$

NWS contributes current severity and hazard mix. Location exposure should come from facilities, employees, insured assets, and critical suppliers. Vulnerability is peril and asset-type specific. Resilience includes insurance, backup capacity, supplier substitution, and adaptation investment. When facility data are absent, the interface says "sector hazard fallback" rather than implying asset-level precision.

4.6 Climate-adjusted FCFF and DCF

The modeled free cash flow is

$$\begin{aligned} FCFF_{i,s,t} = {} & Revenue_{i,s,t} - OperatingCost_{i,s,t} - CarbonCost_{i,s,t} \\ & - Damage_{i,s,t} - AdaptationCapex_{i,s,t}. \end{aligned} \tag{9}$$

The engine applies nonlinear annual paths: orderly transition effects arrive earlier, delayed-transition effects arrive later, and physical losses accelerate toward the horizon. Firm value is

$$V_{i,s} = \sum_{t=1}^{T} \frac{FCFF_{i,s,t}}{(1 + WACC_i)^t} + \frac{FCFF_{i,s,T}(1 + g_{i,s})}{(WACC_i - g_{i,s})(1 + WACC_i)^T}. \tag{10}$$

The discount rate is issuer-specific where market beta is observable and otherwise uses a sector fallback:

$$WACC_i = \frac{E}{V}(r_f + \beta_i ERP) + \frac{D}{V} r_d (1 - \tau_c). \tag{11}$$

The prototype uses disclosed risk-free, equity-risk-premium, debt-cost, tax, leverage, and terminal-growth assumptions. A production model should source debt and capital structure directly from filings and market data.

4.7 Conditional Climate VaR and attribution

For each scenario,

$$ClimateVaR_{i,s} = \frac{V_{i,r} - V_{i,s}}{V_{i,r}}. \tag{12}$$

The model reports each conditional loss, the cross-scenario range, the worst conditional loss, and the mean of the two largest losses as a *model-ensemble tail proxy*. With five stress scenarios and no defensible occurrence probabilities, this proxy is not 95% Expected Shortfall.

Attribution uses exact Shapley values over nine DCF channels: acute physical, chronic physical, country GDP, sector demand, supply chain, carbon cost, energy cost, transition CapEx, and pass-through. The implementation checks that contributions reconcile to total value impact, with a residual near zero.

5 Model-risk sensitivity

The physical loss function is uncertain even when scenario data are official. The platform reruns every scenario with physical damage coefficients scaled to 0.70× and 1.30×:

$$\mathcal{B}_s^{phys} = \left[ClimateVaR_s(0.70\alpha^{phys}),\ ClimateVaR_s(1.30\alpha^{phys}) \right]. \tag{13}$$

This is a transparent sensitivity band, not a confidence interval. It is especially important because NGFS states that the academic paper underpinning parts of Phase V physical-loss estimates was retracted and that an updated methodology is expected in the next long-term scenario release. Therefore, NGFS is treated as one scenario input, not as ground truth.

Future production sensitivity should add alternative damage functions, climate-model ensembles, hazard return periods, adaptation assumptions, WACC shocks, carbon-price paths, demand elasticities, and pass-through ranges.

6 Portfolio construction

6.1 Market return proxy and covariance

The expected-return proxy μ is derived from market information only: 12-month momentum, liquidity, recent return, and volatility control. Climate variables do not receive an assumed return premium. The covariance matrix Σ uses up to 252 aligned daily returns and is annualized; a diagonal variance fallback is labeled when pairwise history is insufficient.

6.2 Robust objective

Within ETF and stock sleeves, the strategic allocation solves the projected-gradient problem

$$\begin{aligned} \max_{w} \quad & \mu^\top w - \lambda w^\top \Sigma w - \kappa \max_{s \in \mathcal{S}} L_s(w) \\ & - \eta (w - w^{mkt})^\top \Sigma (w - w^{mkt}) - \tau \| w - w^{mkt} \|_1, \end{aligned} \tag{14}$$

where

$$L_s(w) = \sum_i w_i \ell_{i,s}. \tag{15}$$

At each iteration the solver identifies

$$s^* = \arg\max_s L_s(w) \tag{16}$$

and uses the loss vector $\ell_{\cdot,s^*}$ as a subgradient. This is materially different from adding each security's independent worst loss, which can overstate or misstate portfolio stress.

The feasible set includes:

$$\mathbf{1}^\top w = 1, \quad 0 \le w_i \le \bar{w}_i, \quad \sum_{i \in sector\ j} w_i \le 35\%,$$
$$\max_s L_s(w) \le B_{scenario}, \quad w^\top p \le B_{physical}, \quad w^\top q \le B_{transition},$$
$$w^\top c \le B_{carbon}, \quad TE(w, w^{mkt}) \le B_{TE}.$$

If the selected universe makes a budget infeasible, the result is marked **Watch**; it is not silently reported as a pass. Broad-market ETFs are modeled as an economy composite rather than defaulting to Technology.

6.3 Tactical overlay

For short-horizon mode, a centered score combines market confirmation and event loss:

$$z_{i,t} = 0.55(M_{i,t} - 0.5) - 0.45E_{i,t}, \qquad \tilde{z}_{i,t} = z_{i,t} - \frac{1}{N}\sum_j z_{j,t}. \tag{17}$$

The overlay is scaled, capped at 250 bps, projected back to each sleeve budget, and then rechecked against position and sector constraints. Long-horizon mode sets this overlay to zero.

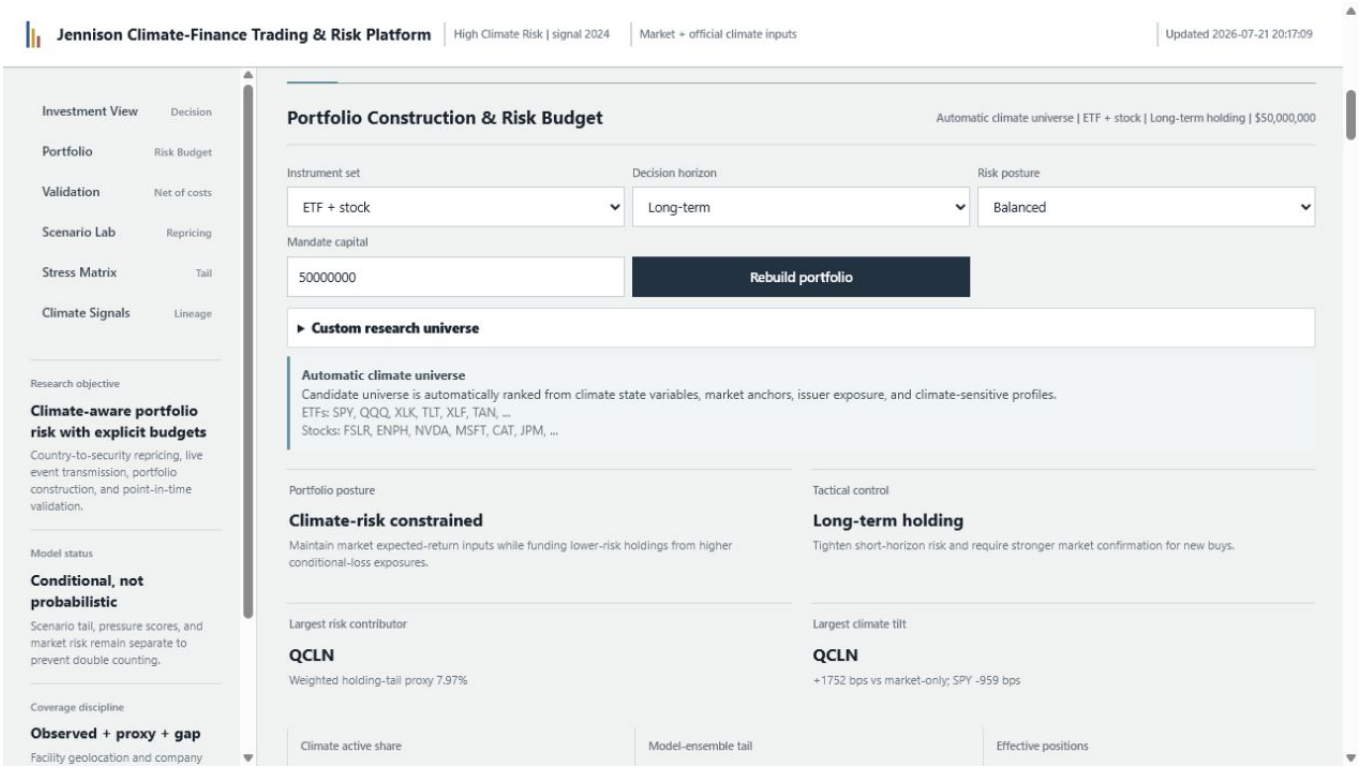


Figure 2: Portfolio Construction and Risk Budget. The interface separates the strategic portfolio, market-only reference, active climate-risk weight, and each feasibility check. Values are an illustrative live run and will change with market data.

7 Walk-forward validation

7.1 Signal timing and execution

For each signal date t, the model forms buy and sell books using only information available through t, then applies the position to $t + 1$ returns. Reported net return is

$$r^{net}_{t+1} = r^{gross}_{t+1} - Turnover_t \times Cost_{trade} - ShortExposure_t \times \frac{Cost_{borrow}}{252}. \tag{18}$$

The benchmark uses the same net risky exposure and cash weight. The three-week window is a monitoring diagnostic; three months is tactical evidence; three years provides broader context but remains conditional on the available universe history.

7.2 Climate-on versus market-only ablation

The central validation estimand is

$$IncrementalClimateValue = Wealth_{FullModel} - Wealth_{MarketOnly}, \tag{19}$$

where both books use the same assets, gross/net exposure, transaction costs, borrow costs, and next-day execution. This design asks whether climate information adds value beyond ordinary market selection rather than comparing unrelated portfolios.

Uncertainty is reported with a five-day moving-block bootstrap and a two-sided sign-permutation test on daily incremental returns. The platform does not call an effect statistically significant merely because total return is positive.

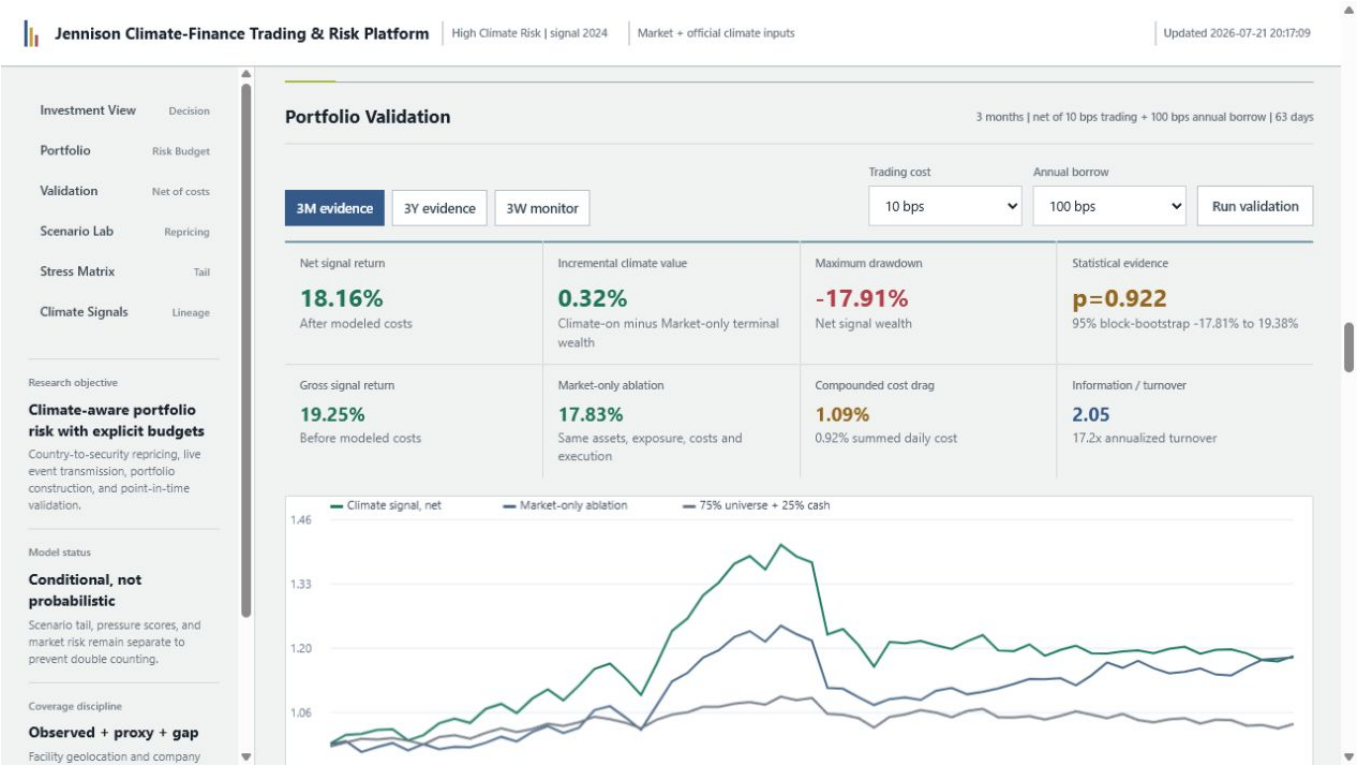


Figure 3: Three-month validation captured on 21 July 2026. In this run the incremental climate value was small and statistically inconclusive; the interface reports the result rather than hiding it.

7.3 Current validation boundary

The present replay is useful for system diagnostics but is not yet production performance evidence. It does not reconstruct a complete archived NWS tape, historical index constituents, delisted securities, ALFRED macro vintages, security-level locate constraints, market impact, financing, or taxes. Until those are loaded, claims should be limited to conditional research evidence.

8 Reading the Scenario and Event Lab

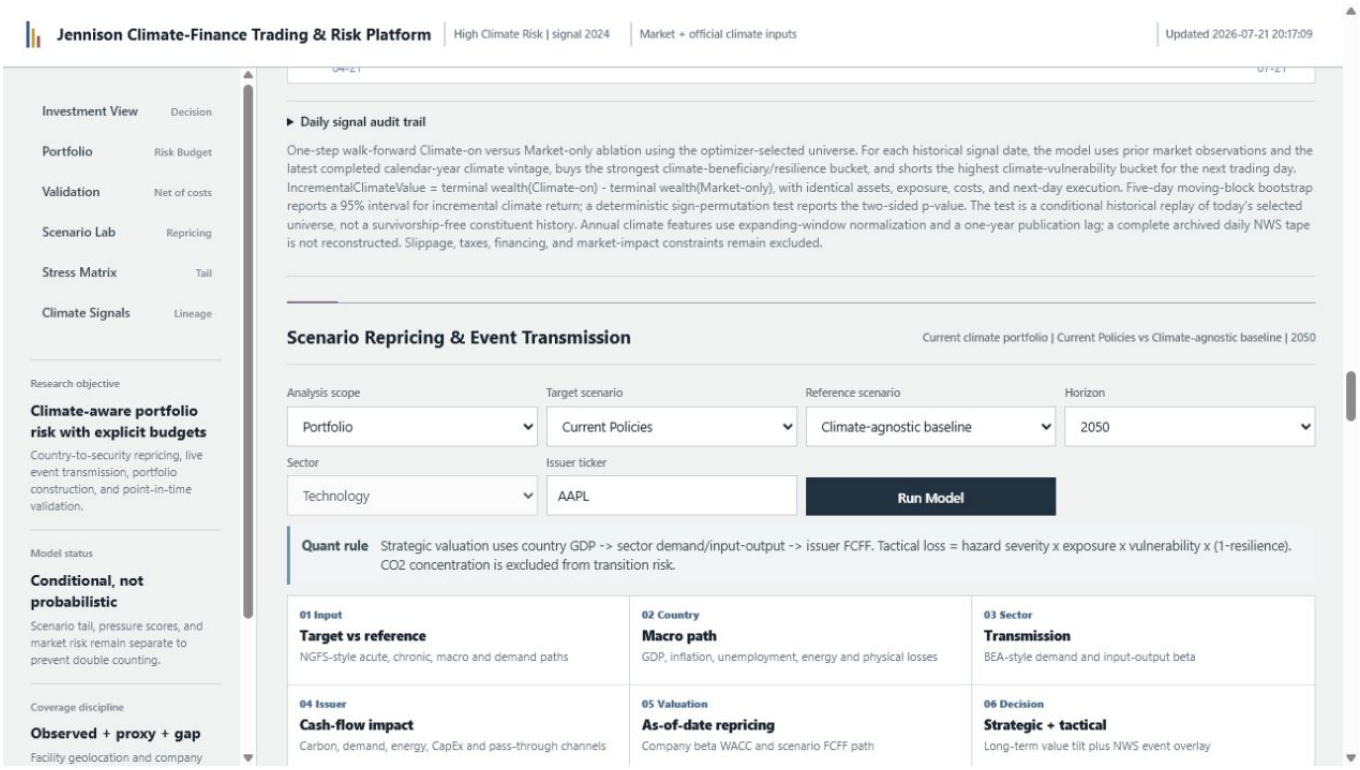


Figure 4: Scenario and Event Lab. Long-term DCF attribution and short-term hazard transmission are displayed together but calculated separately.

The client should read the page in this order:

1. Confirm the target, reference, horizon, and scope.
2. Read total value impact as a conditional as-of-date revaluation, not a probability-weighted forecast.
3. Identify whether acute, chronic, GDP, supply, or transition channels explain the result.
4. Check WACC, terminal growth, issuer fallback quality, and attribution residual.
5. Read NWS event risk separately and confirm that a tactical overlay is zero-sum and capped.
6. Compare the scenario result with the model-risk sensitivity band before making a risk-budget decision.

9 Bloomberg alignment and deliberate differences

The supplied Bloomberg paper describes an obligor-centric, target-versus-reference, as-of-date repricing framework that transmits micro transition, macro transition, acute physical, and chronic physical effects into cash flow and then firm value. It also emphasizes detailed revenue segmentation, supply-chain links, firm-specific WACC, and separate equity/debt allocation.

This prototype adopts the following public and reproducible logic:

- target and reference scenario comparison;
- annual scenario-conditioned cash-flow paths;
- country GDP to sector and issuer transmission;
- separate acute/chronic and micro/macro channels;

- as-of-date DCF using firm/sector-specific WACC;
- portfolio aggregation and contribution analysis;
- explicit data-quality fallbacks.

The prototype deliberately does not claim to reproduce Bloomberg proprietary inputs, TRACT revenue projections, supply-chain data, DRSK default probabilities, Merton-Black-Cox equity/debt allocation, or bond OAS repricing. Its differentiating features are open traceability, a separate real-time event clock, robust portfolio constraints, climate-on/off ablation, and explicit disclosure of unavailable inputs.

10 Controls, tests, and governance

The codebase includes 26 automated methodology tests. Core controls include:

- target equals reference implies zero value impact;
- physical sector downscaling conserves country loss;
- CO_2 changes do not change transition risk;
- transition and GDP components are counted once;
- Shapley contributions reconcile to total value impact;
- NWS overlay is zero-sum and capped, and no hazard mix means no trade;
- missing facility data is labeled as sector fallback;
- portfolio weights, caps, HHI, and risk checks reconcile;
- broad-market ETFs use an economy composite;
- market history is never replaced with simulated returns;
- walk-forward return includes modeled turnover and costs.

Model governance should add versioned parameter files, maker-checker approval for calibration changes, source-file hashes, data-vintage manifests, exception logs, challenger models, and periodic performance decay reviews.

11 Production roadmap

Rank	Workstream	Acceptance criterion	Primary source
1	Point-in-time data store	Every backtest feature has an availability timestamp; archived NWS/NOAA and ALFRED vintages are reproducible	NWS, NOAA, ALFRED
2	Scenario ingestion	Version-pinned annual NGFS variables with unit, model, region, and release metadata	NGFS
3	Macro and supply-chain calibration	BEA GDP-by-industry and requirements tables replace hand-set sector betas	BEA
4	Issuer financials	Revenue, margin, CapEx, debt, tax, and segment data joined by filing date	SEC XBRL
5	Energy transition	Carbon intensity, generation mix, and energy-price channels are issuer/facility specific	EIA, EPA eGRID
6	Physical geolocation	Facilities and critical suppliers mapped to peril layers with validated vulnerability curves	NASA, FEMA, NOAA
7	Portfolio implementation	Prior holdings, turnover, ADV capacity, market impact, borrow, and mandate constraints	Client + market data

Rank	Workstream	Acceptance criterion	Primary source
8	Independent validation	Challenger damage functions, bootstrap stability, attribution, and model-risk committee sign-off	Internal governance

12 Conclusion

The revised model no longer asks a client to trust an abstract climate score. It produces a chain that can be challenged at every step:

source and vintage → economic channel → FCFF path → DCF loss → portfolio constraint → validated action.

The strongest current capability is transparent conditional scenario analysis joined to an auditable portfolio and validation layer. The largest remaining limitation is data granularity: full value will require point-in-time macro vintages, issuer financials, facility locations, product revenue, and supplier/customer links. The correct next step is deeper evidence, not another composite score.

A Parameter and formula ledger

Item	Implemented rule	Governance status
Strategic horizon	Annual paths to 2030, 2040, or 2050	User-selectable
Tactical horizon	1–20 trading days	Separate clock
Physical path shape	$progress^{1.35}$	Sensitivity required
Transition path shape	Orderly 0.70; delayed 2.10; hot house 1.20; fragmented 1.55	Disclosed calibration
Risk-free rate	4.25%	Replace with as-of-date curve
Equity risk premium	4.50%	Governance input
Pre-tax debt cost	5.25%	Replace with issuer spread
Corporate tax rate	21%	U.S. fallback
Single-name cap	20% maximum effective cap	Internal threshold
Sector cap	35%	Hard transfer check
Event overlay	250 bps per security, zero-sum	Hard cap
Scenario probability	None assigned	Conditional stresses
Tail proxy	Mean of two largest conditional losses	Not Expected Shortfall
Physical model risk	0.70–1.30 damage multiplier	Sensitivity, not CI

B Notation

Symbol	Meaning	Symbol	Meaning
i	issuer or instrument	j	sector
c	country	s	climate scenario
h	physical hazard	t	time
w	portfolio weight vector	Σ	annualized return covariance
μ	market-derived expected-return proxy	$L_s(w)$	conditional portfolio loss
$V_{i,s}$	issuer value under scenario	$WACC_i$	issuer discount rate

C References and source notes

References

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